

What is the clinical significance of VHL H115R (c.344A>G) in Von Hippel-Lindau Disease in CIViC?

I will conduct research to find the clinical significance of the VHL H115R (c.344A>G) mutation in Von Hippel-Lindau Disease as documented in the CIViC database. I'll gather and summarize the latest available information.

Worked for 12 minutes >

According to the CIViC (Clinical Interpretations of Variants in Cancer) knowledgebase, the VHL H115R missense variant (c.344A>G, p.His115Arg) is reported as a **rare germ-line variant** in several case reports of von Hippel-Lindau (VHL) disease. These reports collectively suggest that the variant may predispose to VHL-associated tumours, but the evidence remains limited and inconsistent, so CIViC does not yet call it definitively pathogenic.

- **Evidence of predisposition:**

- In a mutational analysis of 16 VHL patients, Wu et al. (2012) found the H115R missense variant in a 45-year-old male patient with central nervous-system hemangioblastomas, renal cell carcinoma and multiple pancreatic cysts. CIViC categorizes this evidence item (EID5772) as a "predisposing" type with "Predisposition" significance, meaning the variant may confer susceptibility but is not proven pathogenic [civcdb.org](#).

- Glavac et al. (1996) reported a German family with VHL type 1 in which two affected relatives shared the H115R variant and developed CNS hemangioblastomas, pancreatic cysts and renal cell carcinoma. CIViC scored this as "supportive" evidence but labeled the significance "Uncertain significance" (EID5377) [civcdb.org](#), reflecting that the variant co-segregates with disease yet remains inconclusive.

- **Uncertain or not-applicable significance:**

- A retrospective cohort study of 291 VHL patients from China found the H115R germ-line variant in a single male patient with CNS hemangioblastoma, renal cell carcinoma and pancreatic lesions; CIViC classified this as "Uncertain significance" [civcdb.org](#).
- Jilg et al. (2012) detected the variant in one of 64 VHL patients with renal involvement; CIViC again categorized this evidence item as "Uncertain significance" [civcdb.org](#).
- Other case reports describing patients with retinal angiomas or hemangioblastomas (EIDs 7748, 8216 and 9913) report the variant but note that the significance is "Not applicable," indicating the evidence is descriptive and cannot determine pathogenicity [civcdb.org](#) [civcdb.org](#) [civcdb.org](#).

Clinical significance in CIViC:

CIViC currently lists no formal Assertion for VHL H115R. Across the seven evidence items, only one study (EID5772) categorizes the variant as a **Predisposition** allele [civcdb.org](#), while three classify it as **Uncertain significance** [civcdb.org](#) [civcdb.org](#) [civcdb.org](#) and the remaining reports label the significance as **Not applicable** [civcdb.org](#) [civcdb.org](#) [civcdb.org](#). Therefore, CIViC views VHL H115R (c.344A>G) as a rare germ-line missense variant with some evidence for disease association but insufficient data to establish pathogenicity; its overall clinical significance for VHL disease remains **uncertain**, and further research and segregation analyses are needed.

Sources

...



Describe a task

+ Agent Sources ▾